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United States Patent	12412869
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chen; Yi-Ling

Electronic structure having first and second thermal conductive materials covering conductive bumps and manufacturing method thereof, and electronic package having electronic structure and manufacturing method thereof

Abstract

An electronic package and a manufacturing method thereof are provided, where the manufacturing method is to dispose an electronic structure with a plurality of conductive bumps and a thermal conductor on its upper surface on a carrier structure via external bumps on its lower surface, so that when reflowing the external bumps, the heat of the heat source joint is conducted from the upper surface of the electronic structure to the external bumps on the lower surface via the thermal conductor, so as to facilitate the heating and reflowing of the external bumps to avoid the problem of non-wetting of the solder.

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Appl. No.:	18/063115
Filed:	December 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240047420 A1	Feb. 08, 2024

Foreign Application Priority Data

TW	111129577	Aug. 05, 2022
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Publication Classification

Int. Cl.: **H01L25/065** (20230101); **H01L21/48** (20060101); **H01L23/00** (20060101); **H01L23/31** (20060101); **H01L23/498** (20060101)

U.S. Cl.:

CPC **H01L25/0655** (20130101); **H01L21/486** (20130101); **H01L23/3128** (20130101);
 H01L23/49816 (20130101); **H01L23/49827** (20130101); **H01L24/73** (20130101);
 H01L24/81 (20130101); H01L2224/73204 (20130101); H01L2224/81815 (20130101)

Field of Classification Search

CPC: H01L (25/0655); H01L (21/486); H01L (23/3128); H01L (23/49816); H01L (23/49827);
 H01L (24/73); H01L (24/81)

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Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a semiconductor device, and more particularly, to an electronic package with through-silicon via (TSV) elements and a manufacturing method thereof.

2. Description of Related Art

(2) In order to ensure the continued miniaturization and multi-functionality of electronic products and communication equipment, semiconductor packaging needs to develop towards miniaturization in order to facilitate the connection of multiple pins. To this end, the industry has developed many advanced packaging process technologies. For example, in advanced packaging process, commonly used packaging types include 2.5D packaging process, fan-out embedded bridge (FO-EB) process, etc.

(3) FIG. 1 is a schematic cross-sectional view showing a manufacturing method of a conventional semiconductor package. As shown in FIG. 1, in the manufacturing method of the conventional semiconductor package, an electronic structure **1a** having at least one conductive TSV **110** is disposed on a routing structure **14** on a carrier board **1b**, wherein a redistribution layer (RDL) **12** electrically connected to the conductive TSV **110** is formed on the surface of a silicon plate body **11**

of the electronic structure **1a**. The RDL **12** includes a dielectric layer **120** and a circuit layer **121** formed on the dielectric layer **120**, and the circuit layer **121** is electrically connected to the conductive TSV **110**. Further, a solder mask layer **13** is formed on the RDL **12**, and parts of the circuit layer **121** are exposed from the solder mask layer **13** to bond to a plurality of conductive bumps **161**.

(4) Furthermore, another solder mask layer **15** may be formed on the silicon plate body **11**, wherein end surfaces of the conductive TSVs **110** are exposed from the solder mask layer **15** to bond to a plurality of solder bumps **122**, and the solder bumps **122** are electrically connected to the conductive TSVs **110** and electrical contact pads **140** of the routing structure **14**, wherein an under bump metallurgy (UBM) layer **17** for connecting the solder bumps **122** can be selectively formed on the end surfaces of the conductive TSVs **110**.

(5) In addition, in the electronic structure **1a**, the plurality of conductive bumps **161** are covered by a protective film **10**.

(6) In the manufacturing method of the conventional semiconductor package, when the electronic structure **1a** is bonded to the routing structure **14**, the solder bumps **122** need to be heated and reflowed. However, during the heating process, a heat source Q provided by a heat source joint **8** of a reflow equipment is deposited on the upper protective film **10** and cannot be effectively transferred to the lower solder bumps **122**, thereby causing the problem of non-wetting of the solder bumps **122**.

(7) Therefore, how to overcome the above-mentioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

(8) In view of the various deficiencies of the prior art, the present disclosure provides an electronic structure, comprising: an electronic body being a semiconductor substrate and having a first surface and a second surface opposing the first surface; a plurality of conductive bumps disposed on the first surface of the electronic body; and a thermal conductor disposed on the first surface of the electronic body and including a first thermal conductive material contacting peripheral surfaces of the plurality of conductive bumps to cover the plurality of conductive bumps, and a second thermal conductive material adjacent to the first thermal conductive material, wherein a thermal conductivity of the second thermal conductive material is greater than a thermal conductivity of the first thermal conductive material.

(9) The present disclosure also provides a method of manufacturing an electronic structure, comprising: providing a semiconductor substrate as an electronic body, wherein the electronic body has a first surface and a second surface opposing the first surface; forming a plurality of conductive bumps on the first surface of the electronic body; forming a first thermal conductive material on a partial region of the first surface of the electronic body, wherein the first thermal conductive material covers the plurality of conductive bumps, and another partial region of the first surface of the electronic body is exposed from the first thermal conductive material; and forming a second thermal conductive material adjacent to the first thermal conductive material on the another partial region of the first surface of the electronic body, wherein the first thermal conductive material and the second thermal conductive material serve as a thermal conductor, wherein a thermal conductivity of the second thermal conductive material is greater than a thermal conductivity of the first thermal conductive material.

(10) In the aforementioned electronic structure and method, a Young's modulus of the first thermal conductive material is less than a Young's modulus of the second thermal conductive material.

(11) In the aforementioned electronic structure and method, a thickness of the first thermal conductive material relative to a peripheral surface of the conductive bump is at least 5 micrometers.

(12) In the aforementioned electronic structure and method, the first thermal conductive material and the conductive bumps are disposed on the first surface of the electronic body in concentric

circles.

(13) In the aforementioned electronic structure and method, a ratio of a layout area of the second thermal conductive material on the first surface to a layout area of the first thermal conductive material on the first surface is greater than or equal to 70%.

(14) The present disclosure further provides an electronic package, comprising: a circuit structure having a first side and a second side opposing the first side; the aforementioned electronic structure disposed on the first side of the circuit structure with the first surface of the electronic body; and a plurality of electronic elements disposed on the second side of the circuit structure and electrically connected to the circuit structure, wherein the electronic structure is electrically connected to the circuit structure via the plurality of conductive bumps to electrically bridge at least two of the plurality of electronic elements.

(15) The present disclosure yet provides a method of manufacturing an electronic package, comprising: providing the aforementioned electronic structure and a circuit structure, wherein the circuit structure has a first side and a second side opposing the first side; disposing the electronic structure on the first side of the circuit structure with the first surface of the electronic body; and disposing a plurality of electronic elements on the second side of the circuit structure, wherein the plurality of electronic elements are electrically connected to the circuit structure, wherein the electronic structure is electrically connected to the circuit structure via the plurality of conductive bumps to electrically bridge at least two of the plurality of electronic elements.

(16) In the aforementioned electronic package and method, the present disclosure further comprises forming a cladding layer covering the electronic structure.

(17) In the aforementioned electronic package and method, the present disclosure further comprises forming a packaging layer covering the plurality of electronic elements.

(18) In the aforementioned electronic package and method, the present disclosure further comprises disposing the electronic structure on a carrier structure with the second surface of the electronic body, wherein the electronic structure is electrically connected to the carrier structure. For example, the electronic structure is mounted to the carrier structure via a plurality of external bumps with the second surface of the electronic body. Alternatively, the present disclosure may comprise forming a plurality of conductive pillars on the carrier structure to electrically connect the carrier structure.

(19) Further, the plurality of conductive pillars surround the electronic structure.

(20) As can be seen from the above, in the electronic package and its manufacturing method and electronic structure and its manufacturing method of the present disclosure, the plurality of conductive bumps are covered by the thermal conductor, so that the second thermal conductive material with higher thermal conductivity can improve the thermal conduction efficiency of the electronic structure, so as to avoid the problem of non-wetting of the solder on the external bumps.

(21) Furthermore, by surrounding and covering the conductive bumps with the first thermal conductive material of softer material, the function of buffering and protecting the conductive bumps can be achieved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package.

(2) FIG. 2A to FIG. 2C-1 are schematic cross-sectional views illustrating a method for manufacturing an electronic structure according to the present disclosure.

(3) FIG. 2C-2 is a schematic top plan view of FIG. 2C-1.

(4) FIG. 2C-3 is a schematic top plan view showing another aspect of FIG. 2C-2.

(5) FIG. 2D is a schematic cross-sectional view showing another aspect of FIG. 2C-1.

(6) FIG. 3A to FIG. 3F are schematic cross-sectional views illustrating a method for manufacturing

an electronic package according to the present disclosure.

(7) FIG. 3G is a schematic cross-sectional view showing another aspect of FIG. 3F.

(8) FIG. 4 is a schematic cross-sectional view of the electronic package according to another embodiment of the present disclosure.

DETAILED DESCRIPTIONS

(9) The following describes the implementation of the present disclosure with examples. Those skilled in the art can easily understand other advantages and effects of the present disclosure from the contents disclosed in this specification.

(10) It should be understood that, the structures, ratios, sizes, and the like in the accompanying figures are used for illustrative purposes to facilitate the perusal and comprehension of the contents disclosed in the present specification by one skilled in the art, rather than to limit the conditions for practicing the present disclosure. Any modification of the structures, alteration of the ratio relationships, or adjustment of the sizes without affecting the possible effects and achievable proposes should still be deemed as falling within the scope defined by the technical contents disclosed in the present specification. Meanwhile, terms such as “on,” “above,” “upper,” “first,” “second,” “one,” “a,” “an” and the like used herein are merely used for clear explanation rather than limiting the practicable scope of the present disclosure, and thus, alterations or adjustments of the relative relationships thereof without essentially altering the technical contents should still be considered in the practicable scope of the present disclosure.

(11) FIG. 2A to FIG. 2C-1 are schematic cross-sectional views illustrating a method for manufacturing an electronic structure 2a according to the present disclosure.

(12) As shown in FIG. 2A, a semiconductor substrate is provided as an electronic body 21, which has a first surface 21a and a second surface 21b opposite to each other, and then a plurality of conductive bumps 211 are formed on the first surface 21a of the electronic body 21. Next, a first thermal conductive material 291 is formed on the first surface 21a of the electronic body 21, so that the first thermal conductive material 291 covers the plurality of conductive bumps 211.

(13) In an embodiment, the electronic body 21 is a semiconductor chip, inside which a plurality of conductive vias 210 connected to the first surface 21a and the second surface 21b are formed, so that the conductive vias 210 are electrically connected to the plurality of conductive bumps 211. For example, the conductive via 210 is a conductive through-silicon via (TSV), and a circuit portion 22 can be formed on the first surface 21a and/or the second surface 21b of the electronic body 21 according to requirements (in the embodiment, the circuit portion 22 is formed on the second surface 21b), wherein the circuit portion 22 includes at least one insulating layer 220 and at least one conductive trace 221 bonded to the insulating layer 220, so that the conductive trace 221 is electrically connected to the conductive via 210. It should be understood that there are various aspects related to the semiconductor device having the conductive via 210, and the present disclosure is not limited to as such.

(14) Furthermore, the conductive bumps 211 are metal pillars such as copper pillars, so that the first thermal conductive material 291 contacts peripheral surfaces 211c and an end surface 211a of each of the conductive bumps 211.

(15) Also, the first thermal conductive material 291 is a dielectric material such as polybenzoxazole (PBO), polyimide (PI), prepreg (PP), or the like.

(16) In addition, a plurality of external bumps 222 may be formed on the second surface 21b of the electronic body 21 (or the circuit portion 22) according to requirements. For example, the external bumps 222 include copper bumps, and the external bumps 222 are electrically connected to the conductive vias 210 (or the conductive traces 221), and a non-conductive film (NCF) can be used as a bonding layer 223 to cover the external bumps 222 according to requirements. It should be understood that there are various aspects related to the external bumps 222, and the present disclosure is not limited to as such.

(17) As shown in FIG. 2B, a patterning process is performed to remove partial materials of the first

thermal conductive material **291**, so that the first thermal conductive material **291** is only formed on a partial region A of the first surface **21a** of the electronic body **21**, such that another partial region B of the first surface **21a** of the electronic body **21** is exposed from the first thermally conductive material **291**.

(18) In an embodiment, a remaining thickness d of the first thermal conductive material **291** relative to the peripheral surface of the conductive bump **211** is at least 5 micrometers, and the first thermal conductive material **291** and the conductive bumps **211** are arranged in concentric circles on the first surface **21a** of the electronic body **21**, as shown in FIG. 2C-2.

(19) As shown in FIG. 2C-1, a second thermal conductive material **292** adjacent to the first thermal conductive material **291** is formed on the first surface **21a** of the electronic body **21**, and the first thermal conductive material **291** and the second thermal conductive material **292** are used as a thermal conductor **29** to form the electronic structure **2a** according to the present disclosure, wherein the thermal conductor **29** is a composite insulating layer, and a thermal conductivity of the second thermal conductive material **292** is greater than a thermal conductivity of the first thermal conductive material **291**.

(20) In an embodiment, a nitride such as silicon nitride (SiN) or other insulating material is used as the second thermal conductive material **292** to fill up the partial region B of the first surface **21a**. For example, the thermal conductivity of the first thermal conductive material **291** is less than 1 W/mK (e.g., PI material), and the thermal conductivity of the second thermal conductive material **292** is 25 W/mK (e.g., SiN material).

(21) Furthermore, a Young's modulus of the first thermal conductive material **291** is less than a Young's modulus of the second thermal conductive material **292**, so that the first thermal conductive material **291** is relatively soft and is regarded as a soft material, and the second thermal conductive material **292** is relatively hard and is regarded as a hard material. For example, the Young's modulus of the first thermal conductive material **291** is 3.3 GPa (e.g., PI material), and the Young's modulus of the second thermal conductive material **292** is 265 GPa (e.g., SiN material).

(22) In addition, although the higher the proportion of the second thermal conductive material **292** is, the better the overall thermal conduction effect will be, but because the peripheral surface **211c** of the conductive bump **211** needs to be covered with a thin film such as the first thermal conductive material **291**, the thickness d of the first thermal conductive material **291** is at least 5 micrometers to achieve the effect of protecting the conductive bump **211**, so the ratio of the arrangement/layout area of the second thermal conductive material **292** on the first surface **21a** to the arrangement/layout area of the first thermal conductive material **291** on the first surface **21a** is preferably greater than or equal to 70% (i.e., ≥ 0.7). For instance, as shown in FIG. 2C-2, when the material of the first thermal conductive material **291** is sufficiently removed in the patterning process of FIG. 2B so that the first thermal conductive material **291** only surrounds the conductive bump **211** with a thickness of 5 micrometers, the layout area of the second thermal conductive material **292** on the first surface **21a** is maximized; alternatively, as shown in FIG. 2C-3, when the material of the first thermal conductive material **291** is not sufficiently removed by the patterning process of FIG. 2B so that only the first surface **21a** between the conductive bumps **211** is exposed from the first thermal conductive material **291**, the layout area of the second thermal conductive material **292** on the first surface **21a** is minimized.

(23) In addition, the second thermal conductive material **292** covers the first thermal conductive material **291** to improve thermal conduction efficiency. Alternatively, in another embodiment as shown in FIG. 2D, the second thermal conductive material **292** is free from covering the first thermal conductive material **291**, and the thermal conductivity of the second thermal conductive material **292** shown in FIG. 2D is less than the thermal conductivity of the second thermal conductive material **292** shown in FIG. 2C-1.

(24) FIG. 3A to FIG. 3F are schematic cross-sectional views illustrating a method for manufacturing an electronic package **2** according to the present disclosure. In an embodiment, the

electronic structure **2a** shown in FIG. 2C-1 is adopted.

(25) As shown in FIG. 3A, a carrier **9** is provided, and at least one electronic structure **2a** and a plurality of conductive pillars **23** are disposed on the carrier **9**.

(26) The carrier **9** is, for example, a plate made of semiconductor material (such as silicon or glass), on which a release layer **90** and a metal layer **91** such as titanium/copper are sequentially formed by, for example, coating, so that a carrier structure **24** is formed on the metal layer **91**, wherein the electronic structure **2a** is disposed on the carrier structure **24** with the second surface **21b** of the electronic body **21** facing the carrier **9**.

(27) In an embodiment, the carrier structure **24** is, for example, a package substrate with a core layer or a coreless package substrate, and the carrier structure **24** includes at least one dielectric layer **240** and a circuit layer **241** bonded to the dielectric layer **240**. For example, the material for forming the dielectric layer **240** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI), prepreg (PP), or the like, and the circuit layer **241** and the dielectric layer **240** may be formed by a redistribution layer (RDL) process.

(28) The electronic structure **2a** is disposed on the circuit layer **241** of the carrier structure **24** via the plurality of external bumps **222** with the second surface **21b** of the electronic body **21**.

(29) In an embodiment, the electronic structure **2a** is fixed on the carrier structure **24** by reflowing the external bumps **222**, wherein the heat source Q provided by the heat source joint **8** of the reflow equipment can be conducted from the first surface **21a** of the electronic body **21** to the external bumps **222** by utilizing the high thermal conductivity of the second thermal conductive material **292**, so as to facilitate the heating and reflow of the external bumps **222**, thereby avoiding the problem of non-wetting of the solder.

(30) Furthermore, if the non-conductive film (NCF) as the bonding layer **223** is not prepared first when the electronic structure **2a** is fabricated, then a bonding layer **223** made of underfill or other materials that are easy to adhere to the dielectric layer **240** may be used to cover the external bumps **222** after the electronic structure **2a** is disposed on the carrier structure **24**.

(31) The conductive pillars **23** are disposed on the carrier structure **24** and are electrically connected to the circuit layer **241**.

(32) In an embodiment, the material for forming the conductive pillar **23** is a metal material such as copper or a solder material. For example, the conductive pillars **23** are formed by electroplating on the circuit layer **241** by exposure and development.

(33) As shown in FIG. 3B, a cladding layer **25** is formed on the carrier structure **24**, so that the cladding layer **25** covers the electronic structure **2a**, the thermal conductor **29** and the conductive pillars **23**, such that a top surface **29a** of the thermal conductor **29**, the end surfaces **211a** of the conductive bumps **211** and end surfaces **23a** of the conductive pillars **23** are exposed from a surface **25a** of the cladding layer **25**.

(34) In an embodiment, the cladding layer **25** is made from an insulating material, such as polyimide (PI), dry film, encapsulating colloid such as epoxy resin, or molding compound. For example, the cladding layer **25** can be formed on the carrier structure **24** by liquid compound, injection, lamination, or compression molding.

(35) Furthermore, the surface **25a** of the cladding layer **25** can be flush with the top surface **29a** of the thermal conductor **29**, the end surfaces **23a** of the conductive pillars **23** and the end surfaces **211a** of the conductive bumps **211** by a leveling process, so that the end surfaces **23a** of the conductive pillars **23** and the end surfaces **211a** of the conductive bumps **211** are exposed from the surface **25a** of the cladding layer **25**. For example, the leveling process removes partial materials of the thermal conductor **29**, partial materials of the conductive pillars **23**, partial materials of the conductive bumps **211** and partial materials of the cladding layer **25** by grinding.

(36) As shown in FIG. 3C, a circuit structure **20** is formed on the cladding layer **25**, so that the circuit structure **20** is electrically connected to the plurality of conductive pillars **23** and the plurality of conductive bumps **211**, wherein the circuit structure **20** has a first side **20a** and a second

side **20b** opposite to each other, so that the electronic structure **2a** is disposed on the first side **20a** of the circuit structure **20** with the first surface **21a** of the electronic body **21**.

(37) In an embodiment, the circuit structure **20** includes at least one insulating layer **200** and at least one redistribution layer (RDL) **201** disposed on the insulating layer **200**, so that the redistribution layer **201** is electrically connected to the plurality of conductive pillars **23** and the plurality of conductive bumps **211**, wherein the outermost insulating layer **200** can be used as a solder mask layer, and the outermost redistribution layer **201** is exposed from the solder mask layer to serve as an electrical contact pad **202**, such as a micro pad (commonly known as μ -pad).

(38) Furthermore, the material for forming the redistribution layer **201** is copper, and the material for forming the insulating layer **200** is a dielectric material such as polybenzoxazole (PBO), polyimide (PI), prepreg (PP), etc., or a solder resist material such as solder mask, graphite (e.g., ink), etc.

(39) As shown in FIG. 3D, a plurality of electronic elements **26** are disposed on the second side **20b** of the circuit structure **20**, and a packaging layer **28** is formed to cover the electronic elements **26**.

(40) In an embodiment, the electronic element **26** is an active element, a passive element, or a combination of the active element and the passive element, where the active element is such as a semiconductor chip, and the passive element is such as a resistor, a capacitor, or an inductor. In one embodiment, the electronic element **26** is, for example, a semiconductor chip such as a graphics processing unit (GPU), a high bandwidth memory (HBM), etc., and the electronic structure **2a** is used as a bridge die, which is electrically connected to the circuit structure **20** via the conductive bumps **211**, thereby electrically bridging at least two electronic elements **26**.

(41) Furthermore, the electronic element **26** has a plurality of conductive bumps **261** such as copper pillars for electrically connecting the electrical contact pads **202** via a plurality of solder materials **260** such as solder bumps. In an embodiment, a UBM layer (not shown) can be formed on the electrical contact pads **202** or the electronic element **26** to facilitate the bonding of the solder materials **260** or the conductive bumps **261**.

(42) In addition, the packaging layer **28** is made from an insulating material, such as polyimide (PI), dry film, encapsulating colloid such as epoxy resin, or molding compound, which can be formed on the circuit structure **20** by lamination or molding. It should be understood that the material for forming the packaging layer **28** may be the same as or different from the material of the cladding layer **25**.

(43) In addition, an underfill **262** can be first formed between the electronic element **26** and the circuit structure **20** to cover the conductive bumps **261** and the solder material **260**, and then the packaging layer **28** can be formed to cover the underfill **262** and the electronic element **26**. Alternatively, in other embodiments, the packaging layer **28** can cover the electronic elements **26** and the conductive bumps **261** at the same time without forming the underfill **262**.

(44) As shown in FIG. 3E, the carrier **9** and the release layer **90** thereon are removed, and then the metal layer **91** is removed to expose the carrier structure **24**.

(45) In an embodiment, when the release layer **90** is peeled off, the metal layer **91** is used as a barrier to avoid damage to the dielectric layer **240** of the carrier structure **24**, so that after removing the carrier **9** and the release layer **90** thereon, the metal layer **91** is removed by etching to expose the circuit layer **241**.

(46) As shown in FIG. 3F, the singulation process is performed along the cutting path S shown in FIG. 3E, and a plurality of conductive elements **27** are formed on the carrier structure **24**, so that the conductive elements **27** are electrically connected to the circuit layer **241** to obtain the electronic package **2**.

(47) In an embodiment, the conductive element **27** includes a metal bump **270** such as copper and a solder material **271** formed on the metal bump **270**. For example, a UBM layer **272** may be formed on the circuit layer **241** to facilitate the bonding of the metal bump **270**. It should be understood

that when the number of the contacts (IOs) is insufficient, the RDL process can still be used to perform build-up layer operations to reconfigure the number and positions of the IOs of the carrier structure **24**.

(48) Furthermore, as shown in FIG. **3G**, partial materials of a packaging layer **38** can be removed by a leveling process, such as grinding, so that a top surface **38a** of the packaging layer **38** is flush with a top surface **26a** of the electronic element **26**, such that the electronic element **26** is exposed from the packaging layer **28**, so as to obtain an electronic package **3**.

(49) Furthermore, the electronic package **2** can be disposed on an electronic device (not shown) such as a package substrate or a circuit board by means of the conductive elements **27**.

(50) Therefore, in the manufacturing method of the present disclosure, the plurality of conductive bumps **211** are covered by the thermal conductor **29**, so that the second thermal conductive material **292** with higher thermal conductivity can improve the thermal conduction efficiency of the electronic structure **2a**, so as to avoid the problem of non-wetting of the solder on the external bumps **222**.

(51) Furthermore, the conductive bump **211** is surrounded and covered by the first thermal conductive material **291** of a relatively soft material to achieve the effect of buffer protection. Therefore, during the leveling process as shown in FIG. **3B**, various problems of uneven interface stress caused by the grinding process can be alleviated by grinding the first thermal conductive material **291** (buffer material).

(52) In addition, please refer to FIG. **4**, the electronic structure **2a** can also be disposed on one side of the circuit structure **20**, and the plurality of electronic elements **26** are disposed on the other side of the circuit structure **20** to obtain an electronic package **4**.

(53) The present disclosure provides an electronic package **2**, **3**, **4**, comprising: an electronic structure **2a**, a plurality of electronic elements **26** and a circuit structure **20**, wherein the electronic structure **2a** includes an electronic body **21**, a plurality of conductive bumps **211** and a thermal conductor **29**.

(54) The electronic body **21** is a semiconductor substrate, which has a first surface **21a** and a second surface **21b** opposite to each other.

(55) The conductive bumps **211** are disposed on the first surface **21a** of the electronic body **21**.

(56) The thermal conductor **29** is disposed on the first surface **21a** of the electronic body **21** and includes a first thermal conductive material **291** contacting the peripheral surfaces **211c** of the plurality of conductive bumps **211** to cover the plurality of conductive bumps **211**, and a second thermal conductive material **292** adjacent to the first thermal conductive material **291**, wherein a thermal conductivity of the second thermal conductive material **292** is greater than a thermal conductivity of the first thermal conductive material **291**.

(57) In one embodiment, a Young's modulus of the first thermal conductive material **291** is less than a Young's modulus of the second thermal conductive material **292**.

(58) In one embodiment, a thickness *d* of the first thermal conductive material **291** relative to the peripheral surface of the conductive bump **211** is at least 5 micrometers.

(59) In one embodiment, the first thermal conductive material **291** and the conductive bumps **211** are disposed on the first surface **21a** of the electronic body **21** in concentric circles.

(60) In one embodiment, a ratio of a layout area of the second thermal conductive material **292** on the first surface **21a** to a layout area of the first thermal conductive material **291** on the first surface **21a** is greater than or equal to 70%.

(61) In one embodiment, the electronic package **2**, **3** further includes a carrier structure **24**, which carries the electronic structure **2a**, so that the electronic structure **2a** is disposed on the carrier structure **24** with the second surface **21b** of the electronic body **21**, such that the electronic structure **2a** is electrically connected to the carrier structure **24**.

(62) Further, the electronic structure **2a** is mounted to the carrier structure **24** via a plurality of external bumps **222** with the second surface **21b** of the electronic body **21**.

(63) Alternatively, the electronic package **2**, **3** further includes a plurality of conductive pillars **23** disposed on the carrier structure **24** to electrically connect the carrier structure **24**. For example, the plurality of conductive pillars **23** surround the electronic structure **2a**.

(64) The circuit structure **20** has a first side **20a** and a second side **20b** opposite to each other.

(65) The electronic structure **2a** is disposed on the first side **20a** of the circuit structure **20** with the first surface **21a** of the electronic body **21**.

(66) The electronic elements **26** are disposed on the second side **20b** of the circuit structure **20** to electrically connect the circuit structure **20**, wherein the electronic structure **2a** is electrically connected to the circuit structure **20** via the plurality of conductive bumps **211** to electrically bridge at least two of the plurality of electronic elements **26**.

(67) In one embodiment, the electronic package **2**, **3**, **4** further includes a cladding layer **25** covering the electronic structure **2a**.

(68) In one embodiment, the electronic package **2**, **3**, **4** further includes a packaging layer **28**, **38** covering the plurality of electronic elements **26**.

(69) To sum up, the electronic package and the manufacturing method thereof of the present disclosure can not only achieve the function of buffer protection, but also improve the thermal conduction efficiency of the electronic structure via the design of the thermal conductor. Therefore, the present disclosure can improve the yield of products.

(70) The foregoing embodiments are provided for the purpose of illustrating the principles and effects of the present disclosure, rather than limiting the present disclosure. Anyone skilled in the art can modify and alter the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of protection with regard to the present disclosure should be as defined in the accompanying claims listed below.

Claims

1. An electronic structure, comprising: an electronic body being a semiconductor substrate and having a first surface and a second surface opposing the first surface; a plurality of conductive bumps disposed on the first surface of the electronic body; and a thermal conductor disposed on the first surface of the electronic body and including a first thermal conductive material contacting peripheral surfaces of the plurality of conductive bumps to cover the plurality of conductive bumps, and a second thermal conductive material adjacent to the first thermal conductive material, wherein the second thermal conductive material is separated from the plurality of conductive bumps by the first thermal conductive material, wherein a thermal conductivity of the second thermal conductive material is greater than a thermal conductivity of the first thermal conductive material.
2. The electronic structure of claim 1, wherein a Young's modulus of the first thermal conductive material is less than a Young's modulus of the second thermal conductive material.
3. The electronic structure of claim 1, wherein a thickness of the first thermal conductive material relative to the peripheral surface of the conductive bump is at least 5 micrometers.
4. The electronic structure of claim 1, wherein the first thermal conductive material and the conductive bumps are disposed on the first surface of the electronic body in concentric circles.
5. The electronic structure of claim 1, wherein a ratio of a layout area of the second thermal conductive material on the first surface to a layout area of the first thermal conductive material on the first surface is greater than or equal to 70%.
6. An electronic package, comprising: a circuit structure having a first side and a second side opposing the first side; the electronic structure of claim 1 disposed on the first side of the circuit structure with the first surface of the electronic body; and a plurality of electronic elements disposed on the second side of the circuit structure and electrically connected to the circuit structure, wherein the electronic structure is electrically connected to the circuit structure via the plurality of conductive bumps to electrically bridge at least two of the plurality of electronic

elements.

7. The electronic package of claim 6, further comprising a cladding layer covering the electronic structure.

8. The electronic package of claim 6, further comprising a packaging layer covering the plurality of electronic elements.

9. The electronic package of claim 6, further comprising a carrier structure carrying the electronic structure, wherein the electronic structure is disposed on the carrier structure with the second surface of the electronic body, and the electronic structure is electrically connected to the carrier structure.

10. The electronic package of claim 9, wherein the electronic structure is mounted to the carrier structure via a plurality of external bumps with the second surface of the electronic body.

11. The electronic package of claim 9, further comprising a plurality of conductive pillars disposed on and electrically connected to the carrier structure.

12. The electronic package of claim 11, wherein the plurality of conductive pillars surround the electronic structure.

13. A method of manufacturing an electronic structure, comprising: providing a semiconductor substrate as an electronic body, wherein the electronic body has a first surface and a second surface opposing the first surface; forming a plurality of conductive bumps on the first surface of the electronic body; forming a first thermal conductive material on a partial region of the first surface of the electronic body, wherein the first thermal conductive material covers the plurality of conductive bumps, and another partial region of the first surface of the electronic body is exposed from the first thermal conductive material; and forming a second thermal conductive material adjacent to the first thermal conductive material on the another partial region of the first surface of the electronic body, wherein the first thermal conductive material and the second thermal conductive material serve as a thermal conductor, wherein the second thermal conductive material is separated from the plurality of conductive bumps by the first thermal conductive material, wherein a thermal conductivity of the second thermal conductive material is greater than a thermal conductivity of the first thermal conductive material.

14. The method of claim 13, wherein a Young's modulus of the first thermal conductive material is less than a Young's modulus of the second thermal conductive material.

15. The method of claim 13, wherein a thickness of the first thermal conductive material relative to a peripheral surface of the conductive bump is at least 5 micrometers.

16. The method of claim 13, wherein the first thermal conductive material and the conductive bumps are disposed on the first surface of the electronic body in concentric circles.

17. The method of claim 13, wherein a ratio of a layout area of the second thermal conductive material on the first surface to a layout area of the first thermal conductive material on the first surface is greater than or equal to 70%.

18. A method of manufacturing an electronic package, comprising: providing the electronic structure of claim 1 and a circuit structure, wherein the circuit structure has a first side and a second side opposing the first side; disposing the electronic structure on the first side of the circuit structure with the first surface of the electronic body; and disposing a plurality of electronic elements on the second side of the circuit structure, wherein the plurality of electronic elements are electrically connected to the circuit structure, wherein the electronic structure is electrically connected to the circuit structure via the plurality of conductive bumps to electrically bridge at least two of the plurality of electronic elements.

19. The method of claim 18, further comprising forming a cladding layer covering the electronic structure.

20. The method of claim 18, further comprising forming a packaging layer covering the plurality of electronic elements.

21. The method of claim 18, further comprising disposing the electronic structure on a carrier

structure with the second surface of the electronic body, wherein the electronic structure is electrically connected to the carrier structure.

22. The method of claim 21, wherein the electronic structure is mounted to the carrier structure via a plurality of external bumps with the second surface of the electronic body.

23. The method of claim 21, further comprising forming a plurality of conductive pillars on the carrier structure to electrically connect the carrier structure.

24. The method of claim 23, wherein the plurality of conductive pillars surround the electronic structure.
